

1. Insertion at the Beginning

Before: [20, 30, 40]

Insert 10

After : [10, 20, 30, 40]

All existing elements are shifted one position to the right.

2. Insertion at a Specific Position (Middle)

Before: [10, 20, 40]

Insert 30 at index 2

After : [10, 20, 30, 40]

Elements from that position onward are shifted right.

3. Insertion at the End

Before: [10, 20, 30]

Insert 40

After : [10, 20, 30, 40]

No shifting required (if space is available).

Time Complexity: $O(1)$

So, in Data Structures, array insertion is generally classified into 3 types:

Insertion at the beginning

Insertion at a specific position (middle)

Insertion at the end